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Peyman Shabani

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CURRENT POSITIONS

Postdoctoral Fellow 2026 – Present
Department of Mechanical and Aerospace Engineering, Carleton University, Ottawa, Canada
Non-Salaried Worker 2021 – Present
Aerospace Research Center, National Research Council Canada, Ottawa, Canada

EDUCATION

Ph.D. in Aerospace Engineering 2021 – 2026
Department of Mechanical and Aerospace Engineering, Carleton University, Ottawa, Canada
Thesis Title: Virtual Testing of Aerospace Composite Materials: Impact and Fatigue
Supervisors: Jeremy Laliberte and Lucy Li

M.Sc. in Mechanical Engineering 2016 – 2019
School of Mechanical Engineering, Iran University of Science and Technology, Tehran, Iran
Thesis Title: Modeling of Matrix Cracking in The Self-Healing Composites
Supervisor: Mahmood M. Shokrieh

B.Sc. in Mechanical Engineering 2012 – 2016
School of Mechanical Engineering, K. N. Toosi University of Technology, Tehran, Iran
Project Title: Design and Analysis of Pressure Vessels in Compliance with ASME Standards

WORK EXPERIENCE

Intern - Airmarket-iART, Alberta, Canada 2025 – Present
Mitacs Business Strategy Internship. Developing computational tools to assist with the safety assessment of a new remotely piloted aircraft system.

Non-salaried Worker - National Research Council Canada, Ottawa, Canada 2021 – Present
Virtual testing of composite materials. Fabrication and testing of composite specimens. Development of NCProCDM user-defined material subroutine for damage prediction in composite materials. Development of the "CompVT: Coupon" Abaqus plugin for the rapid preparation of high-fidelity finite element models. https://compvt.github.io/coupon/. Contribution to the Five Eyes Science and Technology blind prediction challenge on failure analysis of bonded composite structures under static loading.

Research Assistant - Composite Research Laboratory, Iran University of Science and Technology 2016 – 2020
Fabrication of composite specimens. Mechanical testing (static and fatigue). Developing codes for failure life prediction of fiber-reinforced composites.

TEACHING EXPERIENCE

- **Aviation Management and Certification** (AERO 4009) Winter 2025
  Teaching Assistant, Carleton University (Instructor: J. Day)
  Guest lecturer for a session on "Composites Damage Tolerance".
- **Lightweight Structures** (AERO 3101) Winter 2025
  Teaching Assistant, Carleton University (Instructor: M. El Sayed).
- **Capstone Project** (MAAE 4907) Fall 2026, 2025, 2024
  Guest Lecturer, Carleton University (Instructor: S. Ulrich)
  Guest lecturer for a session on "A Practical Introduction to Finite Element Analysis".
- **Composite Materials** (AERO 4608) Fall 2025, 2024, 2023
  Teaching Assistant, Carleton University (Instructor: J. Laliberte)
  Guest lecturer for two sessions on "Finite element analysis of composite materials".
- **Engineering Materials: Strength and Fracture** (MAAE 4102) Fall 2022
  Teaching Assistant, Carleton University (Instructor: A. Artemev).
- **Aerospace Design and Practice** (MAAE 3002) Winter 2024, 2023, 2022
  Teaching Assistant, Carleton University (Instructor: E. Cyr).
- **Mechanics of Solids II** (MAAE 3202) Fall 2021
  Teaching Assistant, Carleton University (Instructor: R. Liu).
- **Continuum Mechanics** Fall 2019
  Teaching Assistant, Iran University of Science and Technology (Instructor: F. Taheri-behrooz).
- **Fatigue in Polymer Composites** Winter 2020, 2019
  Teaching Assistant, Iran University of Science and Technology (Instructor: F. Taheri-behrooz).
- **Machine Design and Practice** Fall 2018
  Teaching Assistant, Iran University of Science and Technology (Instructor: F. Taheri-behrooz).

SKILLS AND RELEVANT COURSES

Programming languages: Python, FORTRAN, MATLAB, C#
Finite element analysis: Abaqus, LS-DYNA, ANSYS
Computer-aided design: SolidWorks, CATIA, AutoCAD
Relevant Coursework: Advanced Composite Materials, Fatigue in Polymer Composites, Damage Mechanics in Composite Materials, Elasticity, Fracture Mechanics, Continuum Mechanics, Finite Element Method, Microstructure and Properties of Materials, Failure Prevention, Additive Manufacturing.

HONORS & AWARDS

- 2026: Canadian Association for Composite Structures and Materials (CACSMA) – Timotei Centea Memorial Award for Best PhD Thesis ($1500)
- 2026: Carleton University – Senate Medal for Outstanding Academic Achievement (for excellence in doctoral work)
- 2025: National Research Council Canada – Aerospace Early Career Network Best Paper Award (Third Place - $250)
- 2024: National Research Council Canada – Aerospace Early Career Network Best Paper Award (Second Place - $350)
- 2021: J.Y. and E.Y. Wong Award for research excellence ($6000)
- 2016: Ranked in the top 1% in the National Graduate Entrance Exam in the field of Mechanical Engineering
- 2012: Ranked in the top 0.5% in the Iranian University Entrance Exam

RESEARCH INTERESTS

Composite Materials, Computational Mechanics, Finite Element Analysis, Fatigue and Fracture, Virtual Testing, Machine Learning.

PUBLICATIONS

Journal Papers

1. **P. Shabani**, L. Li, J. Laliberte. "Progressive fatigue failure analysis of open-hole composite laminates: High-fidelity simulations and an experimental study", International Journal of Fatigue (2026): 109531. https://doi.org/10.1016/j.ijfatigue.2026.109531
2. **P. Shabani**, L. Li, J. Laliberte. "Effects of impactor geometry and multiple impacts on low-velocity impact response and residual compressive strength of fiber-reinforced composite laminates", Composites Part B: Engineering (2025): 112575. https://doi.org/10.1016/j.compositesb.2025.112575
3. **P. Shabani**, L. Li, J. Laliberte. "Low-velocity impact (LVI) and compression after impact (CAI) of Double-Double composite laminates", Composite Structures 351 (2025): 118615. https://doi.org/10.1016/j.compstruct.2024.118615
4. **P. Shabani**, L. Li, J. Laliberte, G. Qi. "Enhanced LaRC05 failure criteria for investigating low-velocity impact on fiber-reinforced composites: an experimental and computational study", Aerospace Science and Technology 155 (2024): 109554. https://doi.org/10.1016/j.ast.2024.109554
5. **P. Shabani**, L. Li, J. Laliberte, G. Qi. "Compression after impact (CAI) failure mechanisms and damage evolution in large composite laminates: High-fidelity simulation and experimental study", Composite Structures 339 (2024): 118143. https://doi.org/10.1016/j.compstruct.2024.118143
6. **P. Shabani**, L. Li, J. Laliberte, G. Qi, D. Rapking, D. Mollenhauer. "High-fidelity simulation of low-velocity impact damage in fiber-reinforced composite laminates using continuum and discrete damage modeling techniques", Composite Structures, 313 (2023): 116910. https://doi.org/10.1016/j.compstruct.2023.116910
7. **P. Shabani**, N. Shabani. "Fatigue life prediction of high-speed composite craft under slamming loads using progressive fatigue damage modeling technique", Engineering Failure Analysis, 131 (2022): 105818. https://doi.org/10.1016/j.engfailanal.2021.105818
8. X. Wu, P. Kanz, H. Mahmoud, J. Millar, **P. Shabani**, J.M. Torres. "Characterization of the Microstructure and Surface Roughness Effects on Fatigue Life Using the Tanaka–Mura–Wu Model", Applied Sciences 11, no. 21 (2021): 9955. https://doi.org/10.3390/app11219955
9. **P. Shabani**, F. Taheri-behrooz, S.S. Samareh-Mousavi, M. M. Shokrieh. "Very high cycle and gigacycle fatigue of fiber-reinforced composites: A review on experimental approaches and fatigue damage mechanisms", Progress in Materials Science (2020): 100762. https://doi.org/10.1016/j.pmatsci.2020.100762
10. **P. Shabani**, M. M. Shokrieh, A. Saeedi. "A novel model to simulate the formation and healing of cracks in self-healing cross-ply composites under flexural loading", Composite Structures, 235 (2020): 111750. https://doi.org/10.1016/j.compstruct.2019.111750
11. **P. Shabani**, M.M. Shokrieh, I. Zibaei. "Effect of the conversion degree and multiple healing on the healing efficiency of a thermally reversible self-healing polymer", Polymers for Advanced Technologies, 30 (2019), pp. 2906-2917. https://doi.org/10.1002/pat.4723
12. **P. Shabani**, F. Taheri-behrooz, S. Maleki, M. Hasheminasab. "Life Prediction of a notched composite ring using progressive fatigue damage models", Composites Part B: Engineering, 165 (2019), pp. 754-763. https://doi.org/10.1016/j.compositesb.2019.02.031

Book Chapters

1. **P. Shabani**, M. M. Shokrieh. "Modeling of crack self-healing in thermally remendable fiber-reinforced composites" In Multifunctional Epoxy Resins: Self-Healing, Thermally and Electrically Conductive Resins, pp. 239-276. Singapore: Springer Nature Singapore, 2023. https://doi.org/10.1007/978-981-19-6038-3_9

Conference Papers

1. **P. Shabani**, M. Lanzo, L. Li, J. Laliberte. "Numerical Investigation of Hybrid Thin-Ply Laminates for Matrix Cracking and Delamination Suppression in Linerless Cryogenic Hydrogen Tanks." In CANCOM 2026 – 14th Canadian International Conference on Composite Materials, Vancouver, Canada. 2026. https://www.cacsma.ca/wp-content/uploads/2026/07/CANCOM2026_paper_105.pdf
2. L. Li, **P. Shabani**, J. Laliberte. "Virtual Testing of Progressive Failure of Composite Joints with Localized Debonding." In NATO Science and Technology Organization (STO) Applied Vehicle Technology (AVT), Bordeaux, France. 2026. https://publications.sto.nato.int/publications/STO%20Meeting%20Proceedings/STO-MP-AVT-412/MP-AVT-412-06.pdf
3. M. Lanzo, **P. Shabani**, L. Li, J. Laliberte. "Design Optimization for Automated Fibre Placement Manufacturing of Linerless Hydrogen Composite Pressure Vessels." In CANCOM 2026 – 14th Canadian International Conference on Composite Materials, Vancouver, Canada. 2026. https://www.cacsma.ca/wp-content/uploads/2026/07/CANCOM2026_paper_109.pdf
4. L. Li, A. Pilkington, N. Du, W. Qian, A. Islam, M. Lanzo, **P. Shabani**, J. Laliberte. "Enhancing Permeability Resistance of Composite Cryogenic Hydrogen Storage for Sustainable Aviation." In CANCOM 2026 – 14th Canadian International Conference on Composite Materials, Vancouver, Canada. 2026. https://www.cacsma.ca/wp-content/uploads/2026/07/CANCOM2026_paper_30.pdf
5. **P. Shabani**, L. Li, J. Laliberte. "Progressive Failure Analysis of Bonded Composite Specimens Under T-Pull and Side-Bend Loadings Using Integrated Discrete-Continuum Damage Models." In AIAA SCITECH 2026 Forum, p. 0448. 2026. https://doi.org/10.2514/6.2026-0448
6. **P. Shabani**, L. Li, J. Laliberte. "High-fidelity simulation of low-velocity impact (LVI) and compression after impact (CAI) in fiber-reinforced composite laminates using Abaqus.", SIMULIA Americas Users Conference, Novi, Michigan, USA.
7. M. Lanzo, **P. Shabani**, L. Li, J. Laliberte. "Structural integrity assessment of type IV hydrogen pressure vessels: Drop testing and fatigue damage modelling", CSME 2025 – Canadian Society for Mechanical Engineering, Montreal, Canada. https://doi.org/10.82417/zww3-6h82
8. **P. Shabani**, L. Li, J. Laliberte. "Low-velocity impact (LVI) and compression after impact (CAI) of Double-Double composite laminates", CANCOM2024 – Canadian International Conference on Composite Materials, Waterloo, Canada, August 2024. https://www.cacsma.ca/wp-content/uploads/2024/10/CANCOM_2024_paper_89.pdf
9. L. Li, J. Vallée, **P. Shabani**, J. Laliberté, A. Dondish. "Understanding the effect of global buckling on composites damage tolerance", CANCOM2024 – Canadian International Conference on Composite Materials, Waterloo, Canada, August 2024. https://www.cacsma.ca/wp-content/uploads/2024/10/CANCOM_2024_paper_90.pdf
10. **P. Shabani**, L. Li, J. Laliberte, G. Qi. "Balancing high-fidelity and model efficiency of damage prediction of composite structures", 20th International Conference on Experimental Mechanics, Porto, Portugal, July 2023. https://paginas.fe.up.pt/~icem20/proceedings_icem20/data/papers/19892.pdf
11. **P. Shabani**, L. Li, J. Laliberte, G. Qi. " Low-velocity impact damage simulation using cohesive zone model and LaRC05 criterion with efficient search algorithm", 23rd International Conference on Composite Materials, Belfast, Northern Ireland, August 2023.
12. L. Li, **P. Shabani**, J. Laliberte, G. Qi. "Virtual testing of low-velocity impact response of a composite laminate - from analytical to high-fidelity modelling". ICAF 2023 – the 38th Conference and 31st Symposium of the

International Committee on Aeronautical Fatigue and Structural Integrity, Delft, The Netherlands, July 2023. **(Received Best Poster Award: €500)** https://www.icaf.aero/icaf2023/proceedings/display_manuscript/126.htm

13. **P. Shabani**, G. Qi, L. Li, J. Laliberte. "Modeling low-velocity impact damage behavior of fiber-reinforced composites using combined continuum and discrete damage modeling techniques". CANCOM2022 – Canadian International Conference on Composite Materials, New Brunswick, Canada, July 2022.
14. **P. Shabani**, L. Li, J. Laliberte, G. Qi. "A benchmarking evaluation of finite element modeling strategies for adhesively-bonded composites using the cohesive zone modeling and virtual crack closure technique". ICFC8 - The 8th International Conference on the Fatigue of Composites, Padua, Italy, June 2021.
15. **P. Shabani**, M.M. Shokrieh, I. Zibaei. "Effect of the conversion degree on the healing efficiency of a novel thermally remendable polymer". CCFA-6 - Composites: Characterization, Fabrication & Application, Tehran, Iran, December 2018.

Technical Reports

1. **P. Shabani**, L. Li, J. Laliberte. "LTR-SMM-2026-XXXX: Artificial Intelligence and Machine Learning for Composite Materials and Structures", National Research Council Canada, 2026. (In Preparation)
2. **P. Shabani**, M. Lanzo, L. Li, J. Laliberte. "LTR-SMM-2026-XXXX: Stress and failure analysis of composite hydrogen storage tanks", National Research Council Canada, 2026. (In Preparation)
3. **P. Shabani**, G. Qi, L. Li, J. Laliberte. "LTR-SMM-2023-0026: An Effective Modelling and Simulation Strategy for Predicting Compression After impact (CAI) Strength of Composite Laminates", National Research Council Canada, 2023.
4. **P. Shabani**, G. Qi, L. Li, J. Laliberte. "LTR-SMM-2023-0125: Development of an accurate composite laminate damage modelling methodology with improved LaRC05 failure criteria", National Research Council Canada, 2023.
5. **P. Shabani**, G. Qi, L. Li, J. Laliberte. "LTR-SMM-2022-0011: Simulation of low-velocity impact damage behavior of fiber-reinforced composite laminates using discrete and continuum damage modelling techniques", National Research Council Canada, 2022.

ACADEMIC SERVICES

PEER-REVIEW (2023-PRESENT) Composites Part B: Engineering (Elsevier), Composite Structures (Elsevier), Composites Part C: Open Access (Elsevier), International Journal of Mechanical Sciences (Elsevier), Thin-walled Structures (Elsevier), Advanced Composites and Hybrid Materials (Springer), Applied Composite Materials (Springer), Discover Materials (Springer), Polymer Composites (Wiley), International Journal of Damage Mechanics (Sage), Processes (MDPI).

CONTACT & LINKS

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- **LinkedIn**: https://ca.linkedin.com/in/peyman-shabani
- **ResearchGate**: https://www.researchgate.net/profile/Peyman-Shabani-2

REFERENCES

- **Jeremy Laliberte** (PhD thesis supervisor), Professor, Department of Mechanical and Aerospace Engineering, Carleton University, Canada. Email: Jeremy.laliberte@carleton.ca
- **Lucy Li** (PhD thesis co-supervisor), Principal Research Officer, Aerospace Research Centre, NRC Canada. Email: Chun.li@nrc-cnrc.gc.ca